Gravisonde Architecture Overview

flowchart TB subgraph External Judge[Judge's Browser] Camera[Camera / RTSP / File] end subgraph AWS VPC[VPC 2 AZ] EC2[EC2 c8g.xlarge Graviton4
COOL AMI OpenCV 5] S3[S3 Benchmark Artifacts] CW[CloudWatch Logs + Metrics] ALB[ALB HTTPS] end subgraph LocalDev Dashboard[Next.js 16 Dashboard
App Router + SSE] Pipeline[OpenCV 5 Pipeline
YOLO11n + ALIKED/LightGlue] LS[LocalStack
CloudWatch/S3 mock] end Camera --> EC2 EC2 --> CW EC2 --> S3 ALB --> Dashboard Dashboard --> Judge Pipeline --> Dashboard Pipeline --> LS style EC2 fill:#0ea5e9,stroke:#6366f1 style Dashboard fill:#6366f1,stroke:#0ea5e9

Layers

1. Camera → EC2 Graviton4 COOL AMI
2. OpenCV 5 DNN + Features2D
3. Telemetry → CloudWatch + S3
4. Next.js 16 Dashboard SSE
5. Judge browser live view